

Real Time In-DAW Singing Synthesis

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I. INTRODUCTION

SPEECH synthesis— that is, the task of creating voice-like audio using technology, has fascinated researchers for years. The plugin presented here makes it possible to do speech synthesis directly in a digital audio workstation (DAW). It does so by defining a number of different synthesizers, which are each given access to the audio buffer one after the other, as is shown in the signal flow diagram in Figure 1.

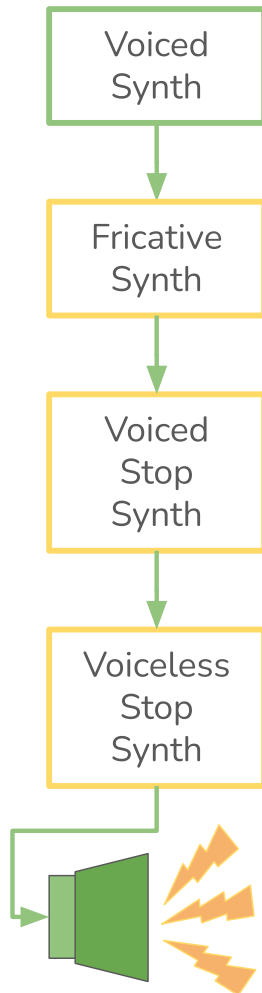


Fig. 1. A signal flow diagram of the entire plugin.

In the following discussion, each of these synthesizer layers will be examined in detail. Note that throughout this discussion, as well as in the presets provided with the plugin, the

international phonetic alphabet (IPA) is used. More information about the IPA, as well as example recordings of sounds, is given in [1]¹.

II. THE VOWEL SYNTHESIZER

At the heart of this product is the vowel synthesizer, referred to in the code base as VoicedSynth. This synthesizer is based on the source-filter model of speech, which has been used in speech synthesis contexts for many years. As the name suggests, this model separates speech into two independent components: the source, which is a model of the vibration of the vocal folds, and the filter, which is a model of the resonance of the oral cavity or vocal tract. This synthesizer can be used to synthesize vowel sounds within the midi note range [C1, F3]. A signal flow diagram of this synthesizer can be seen in Figure 2.

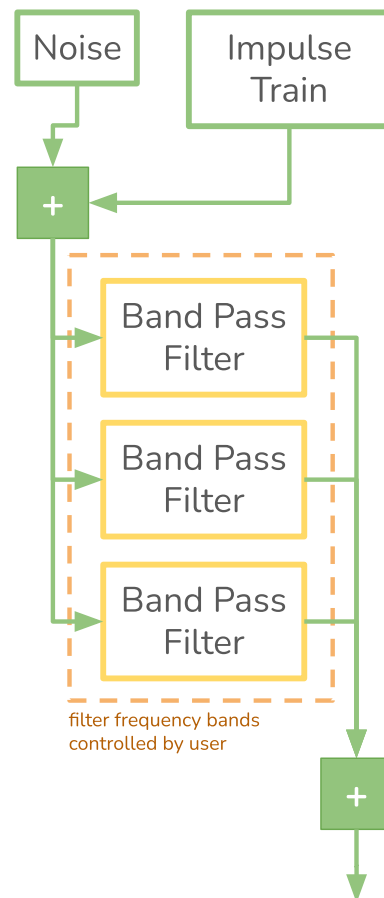


Fig. 2. A signal flow diagram of the VoicedSynth component of the plugin.

¹<https://www.seeingspeech.ac.uk/ipa-charts/>

A. The Liljencrants-Fant Model of the Glottis

During voiced speech, the vocal folds slap together hundreds of times per second. In source-filter speech synthesis, this is modeled by some variant of a pulse train. Following work done by researchers at The Centre for Speech Technology Research at the University of Edinburgh, the Liljencrants-Fant model (LF model) is used to represent the glottal source in this project [2]. This model is used because it has been extensively studied in the context of not only speech synthesis, but singing synthesis as well [3]. Random noise was mixed with the glottal signal as shown in [2], and calculations were performed as outlined in [2], using values for parameters from [3]. Notably, notes under 80 Hz use parameters meant to elicit a creakier voice, and notes over 276 Hz use parameters meant to elicit a breathier voice, in order to mimic the acoustic effects of a human voice singing at the boundaries of its comfortable range, as suggested by [3], [4].

B. The Vocal Tract

In speech, the shape of your mouth determines the predominant frequencies of a voiced sound, which is how we tell vowels apart. In this implementation of the source-filter model, three band pass filters are run in parallel, which capture the first three formants, which are the most important frequency bands for differentiating vowels.

The central frequencies of these three bands are the only controllable parameters of the synthesizer. There are two modes by which the user can control these parameters. The first is using proxy parameters that have been thoughtfully designed based on physical properties of articulatory phonetics. These parameters, called closure, advancement, and rounding, serve to maximize accessibility of the synthesizer. Closure represents how closed the mouth is, as well as how raised the tongue is, with a value of 0 representing a very closed mouth and a value of 10 representing a wide open mouth. This parameter has a direct positive correlation with the frequency of the first formant, as seen in chapter 9 of [5]. The second of these parameters, advancement, represents how far forward the tongue is in the mouth, with 0 representing a far back tongue and 10 representing a far forward tongue. This parameter correlates strongly with the difference between formant 1 and formant 2, as seen in [5], [6]. Lastly, rounding represents whether or not the lips are rounded, with 0 representing no rounding and 10 representing very rounded lips. In the physical world, the acoustic effect of rounding is vaguely defined, but in this project rounding has the effect of lowering the frequency of formant 2, and decreasing the amplitude of the component of the signal passing through the third formant, as suggested by [7].

The second mode by which these formant frequencies can be modified is by manual adjustment, which can be enabled by the checkbox labeled "manual formant control".

III. THE CONSONANT SYNTHESIZERS

Alongside the vowel synthesizer discussed above, the project also includes three distinct consonant synthesizers. 11 stop and fricative consonants can be synthesized, including the

voiced stops b, d, and g, the aspirated voiceless stops p^h , t^h , and k^h , and the fricatives f/v, θ/δ , s/z, $\int/\zeta$, and h.

A. Fricatives

The midi notes G4, A4, B4, C5, D5 correspond to the fricatives f/v, θ/δ , s/z, $\int/\zeta$, and h. respectively. These consonants are synthesized by applying a low-pass filter with a cutoff frequency of 1,100 Hz to any voiced speech, and then producing noise that is filtered with a band-pass filter, with a frequency and Q meant to elicit the correct sound, which is discussed in [5].

Since these consonants are created in this way, there is no need to differentiate between voiced and voiceless fricatives (e.g. z and s), as the presence or absence of any voicing will do so implicitly. A signal flow diagram of this process is shown in Figure 3.

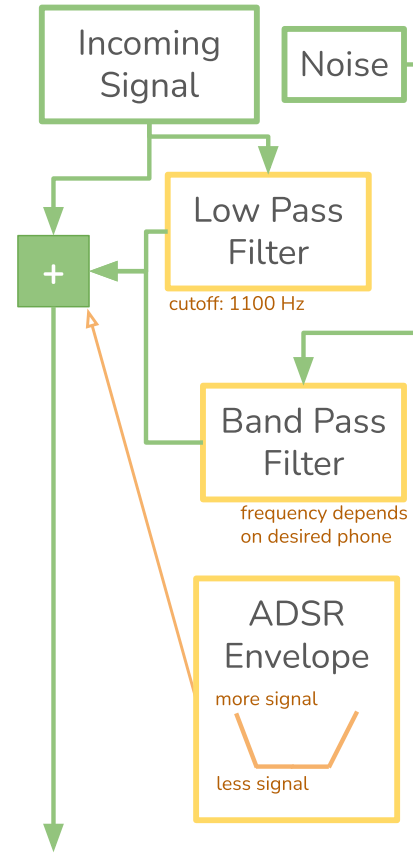


Fig. 3. A signal flow diagram of the FricativeSynth component of the plugin.

B. Voiced Stops

The midi notes C4, D4, F4 correspond to voiced stops b, d, and g, respectively. These stops are represented by applying a low-pass filter with a very low cutoff frequency of 80 Hz, to simulate the closure of the oral cavity, as well as slight modifications of the formant frequencies, as discussed in [5]. A signal flow diagram of this process is shown in Figure 4.

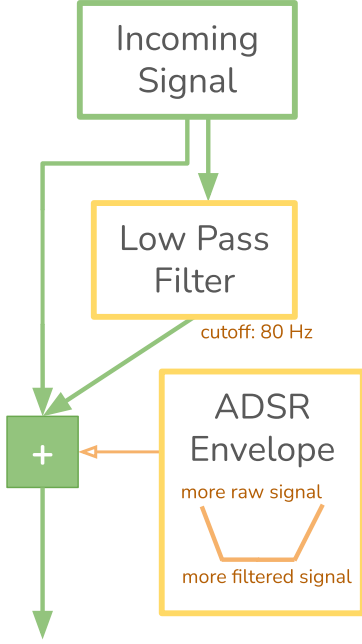


Fig. 4. A signal flow diagram of the VoicedStopSynth component of the plugin.

C. Voiceless Stops

The midi notes C4, D4, E4 correspond to aspirated voiceless stops p^h , t^h , and k^h , respectively. These stops are represented very similarly to the voiced stops discussed above, with two key differences. Firstly, when one of these stops are in effect, all other sounds produced by the synthesizer are stopped completely, to produce a truly voiceless stop. Secondly, and most importantly, aspiration is synthesized after releasing one of these consonants. This aspiration is created by synthesizing filtered noise at a certain formant frequency, and delaying the onset of voicing (i.e. slowly fading in any voiced sounds), as described in [5]. A signal flow diagram of this process is shown in Figure 5.

IV. EXAMPLE USE CASES

Three example use cases of this project will be presented here, along with demo audio to showcase them in action.

A. Placeholder Choir

From the perspective of a choral arranger, this project could be a useful tool to listen to what an in-progress choral piece would sound like, without the requirement of a real singer or singers. This project is particularly useful for this task because not only does it synthesize notes and vowel formants, but it synthesizes consonants as well, which would be useful to hear how potential lyrics would sound with a real choir. Further, a choir could record the final piece while listening to that placeholder audio, which could help with lining up consonants, which is a necessity in high quality choir recording.

Within the assignment submission, Choir.wav provides an example of this use case. Within this excerpt, music from

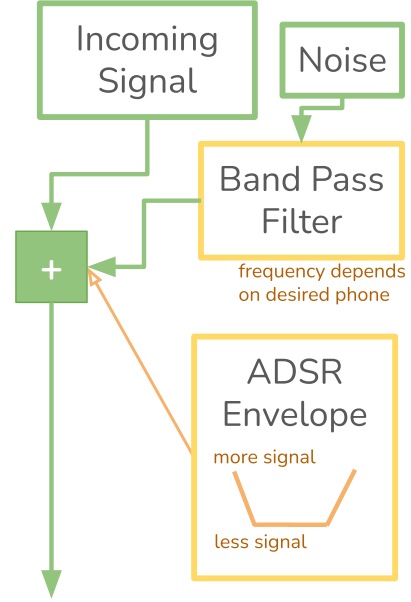


Fig. 5. A signal flow diagram of the VoicelessStopSynth component of the plugin.

Sky Above by Jacob Collier is synthesized with different consonants, to demonstrate various uses within this space.

B. Perception Study

From the perspective of a linguist, this project could be a useful tool when performing studies about perception. A number of studies have been performed where, for example, the threshold at which point a listener perceives one vowel or another is examined (as in [8]). This project provides a user-friendly way to synthesize vowels with exact formant frequencies, allowing for the production of synthesized vowels along a given formant continuum.

Within the assignment submission, Linguistics.wav provides an example of this use case. At the beginning of this file, the word "bead" is synthesized by the plugin. At the end, the word, "bad" is synthesized. Between these two, the consonants b and d repeatedly surround a synthesized vowel that is between the vowel in "bead" (i) and the vowel in "bad" (æ).

C. Artistic Expression

From the perspective of a producer, this project could be a useful tool when making a piece of music, because it can be used to elicit an interesting, voice-like sound without needing to put a voice talent in front of a microphone.

Within the assignment submission, Hype.wav provides an example of this use case. Within this excerpt, the words "Go" and "Hey" are repeatedly synthesized by the plugin, in an attempt to make the beat more cool.

V. LIMITATIONS

One noted limitation of the current product is that the noise bursts within the aspirated consonants (p^h , t^h , and k^h)

discussed in Section III-C do not sound very realistic. Future work on this project will further tweak the onset time of these bursts, as well as the frequency and Q value of the band-pass filter that is filtering the noise, in order to better match real productions of these sounds.

VI. CONCLUSION

REFERENCES

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